AP 18.3-19. 10.12.2025

Erythrocytes → are biconcave disks
↳ they are plump at their periphery & very thin center

Since they lack most organelles → more interior spaces of for hemoglobin molecules

greater surface area -> biconcave for gas exchange

O_2 diffuse through cappillaries - to reach cells
CO_2 is picked up.

spectrin allows the structure to bend on itself → & then rebound. They can also form a rouleaux

Hemoglobin → large molecule made of proteins & iron. It consists of 4 folded chains of proteins called
Globin → designated alpha 1, 2
beta 1, 2

Each globin molecule is bound a to red pigment molecule → (Heme) which contains an ion of Iron or Fe^{2+}

→ Each iron ion in the Heme can bind to 1 O_2 molecule; therefore each hemoglobin molecule can transport four O_2 molecules

One RBC may contain 300 million hemoglobin molecules -> can bind to & transport 1.2 billion O_2 molecules